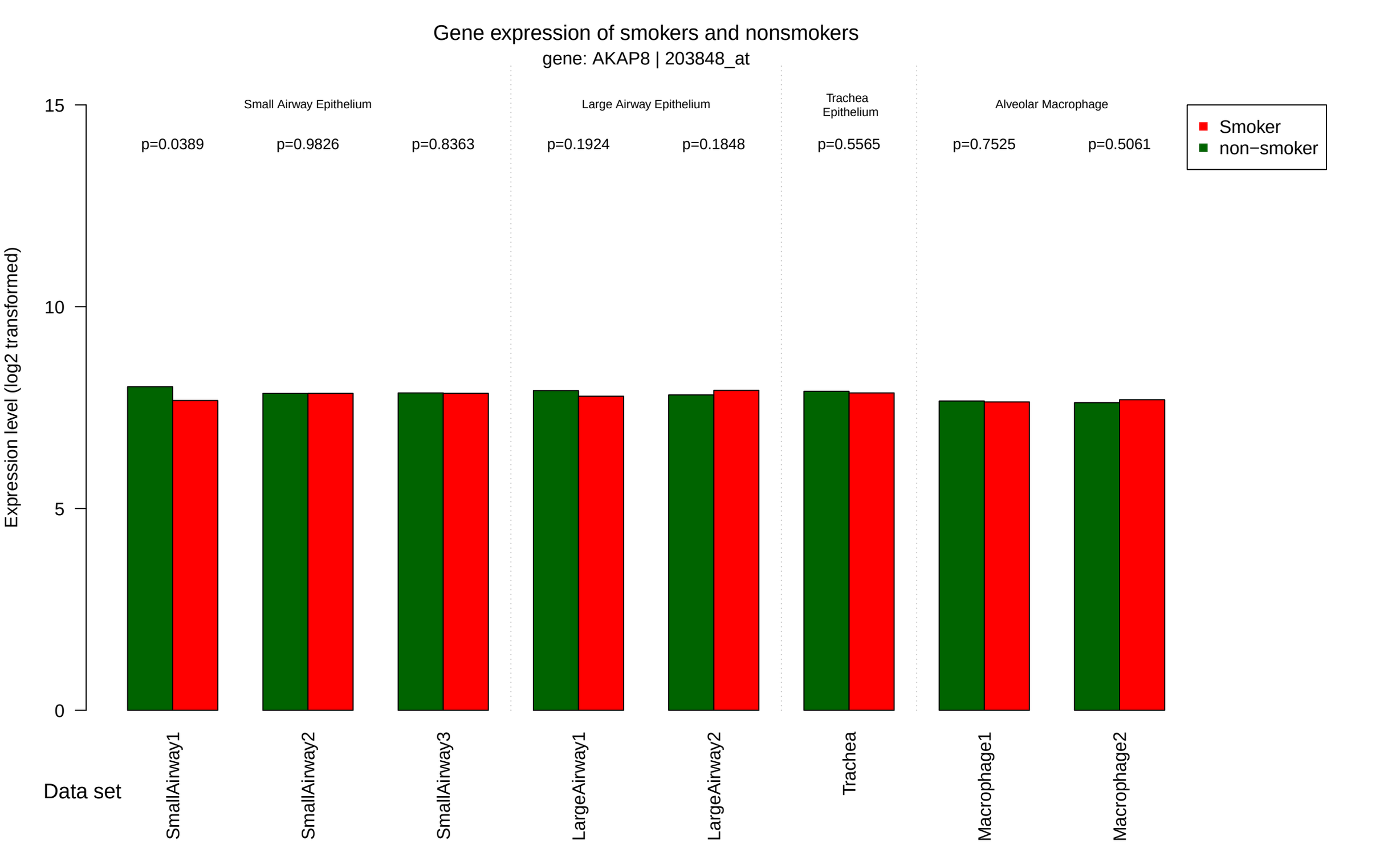
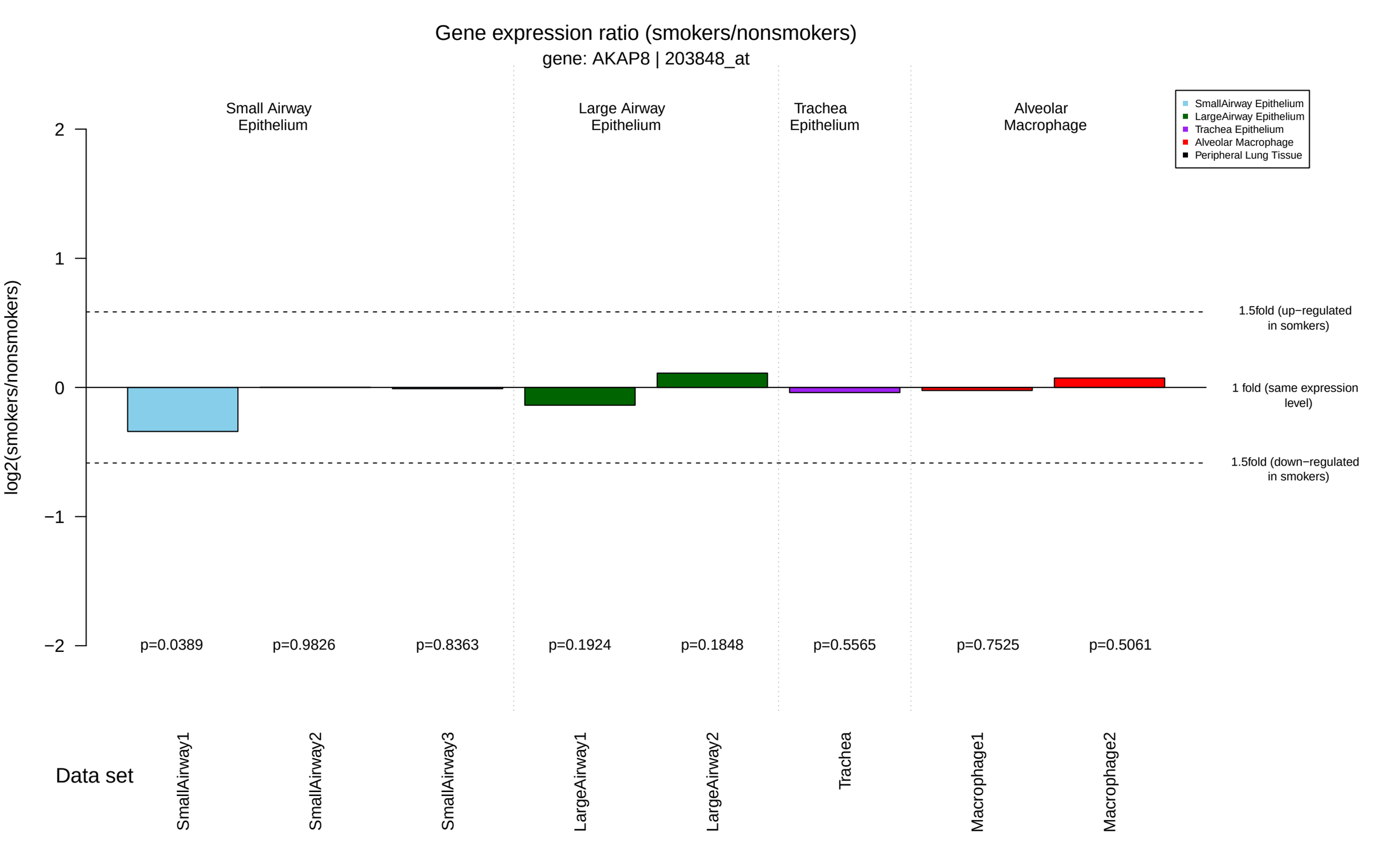
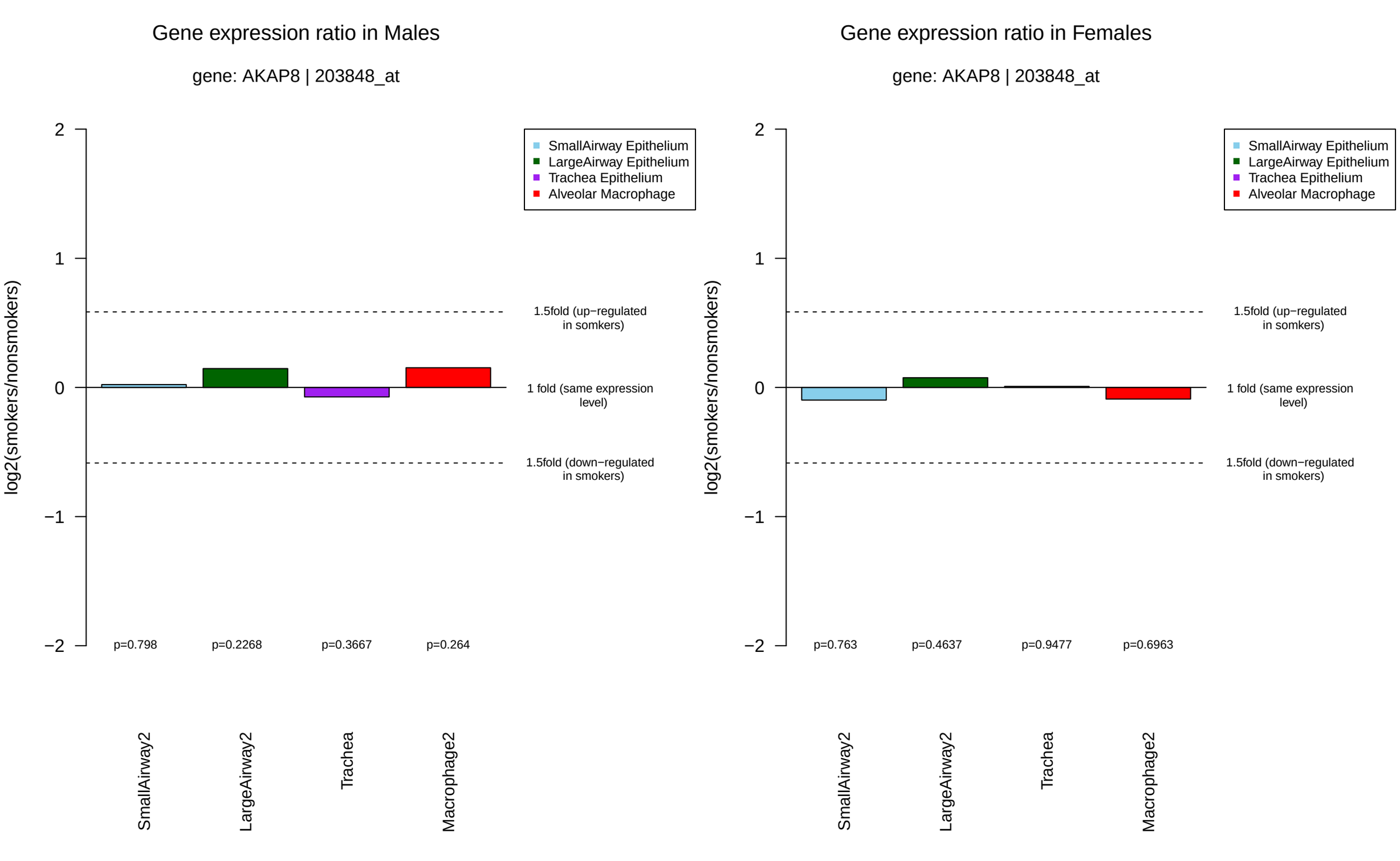
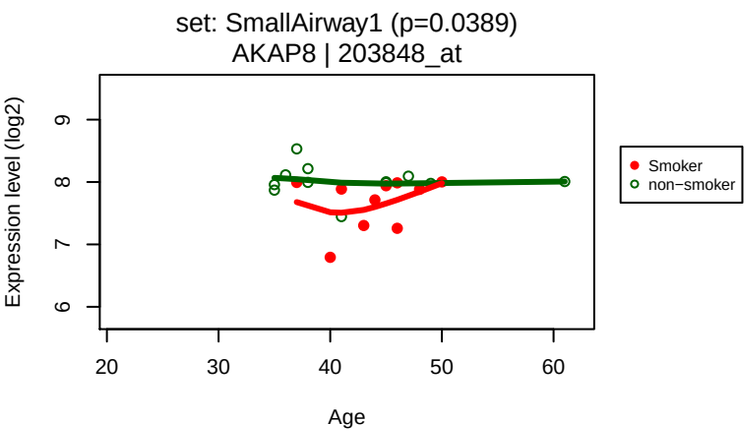
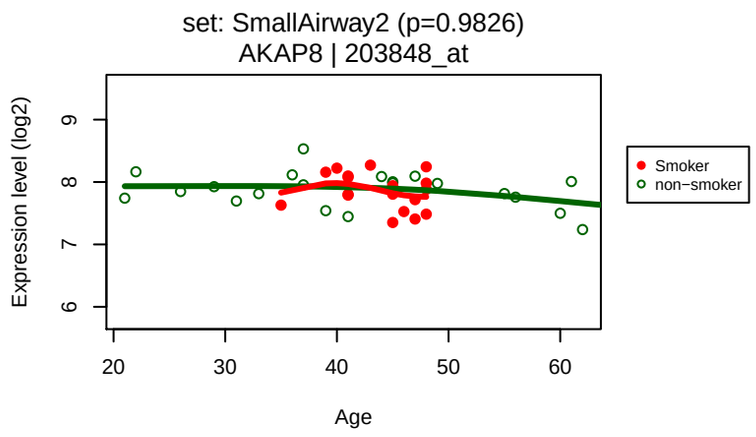
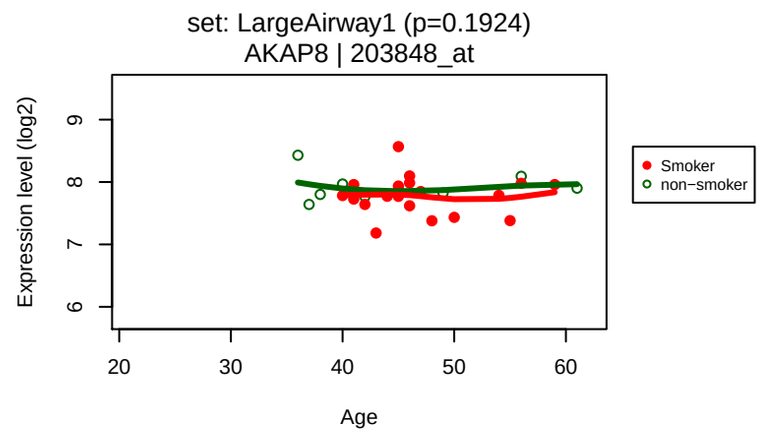

Gene expression of smokers and nonsmokers
gene: AKAP8 | 203848_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0389
p=0.9826
p=0.8363
p=0.1924
p=0.1848
p=0.5565
p=0.7525
p=0.5061
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

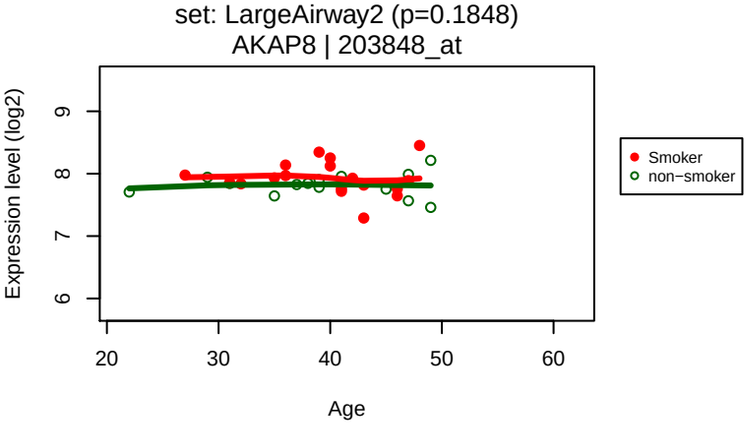

Gene expression ratio (smokers/nonsmokers)
gene: AKAP8 | 203848_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0389
p=0.9826
p=0.8363
p=0.1924
p=0.1848
p=0.5565
p=0.7525
p=0.5061
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

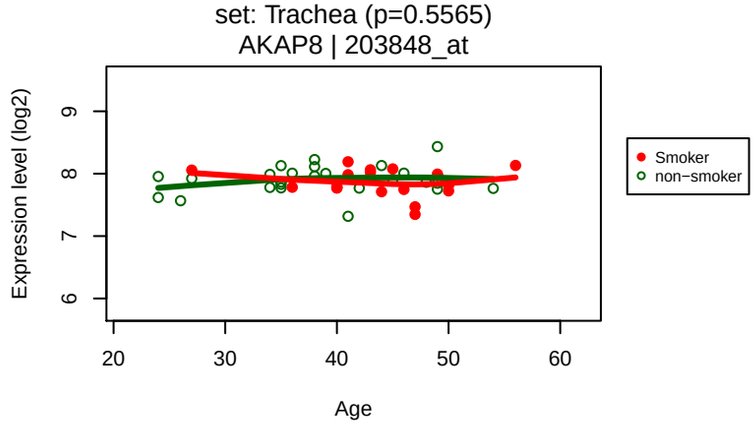

Gene expression ratio in Males
gene: AKAP8 | 203848_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.798
p=0.2268
p=0.3667
p=0.264
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP8 | 203848_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.763
p=0.4637
p=0.9477
p=0.6963
SmallAirway2
LargeAirway2
Trachea
Macrophage2

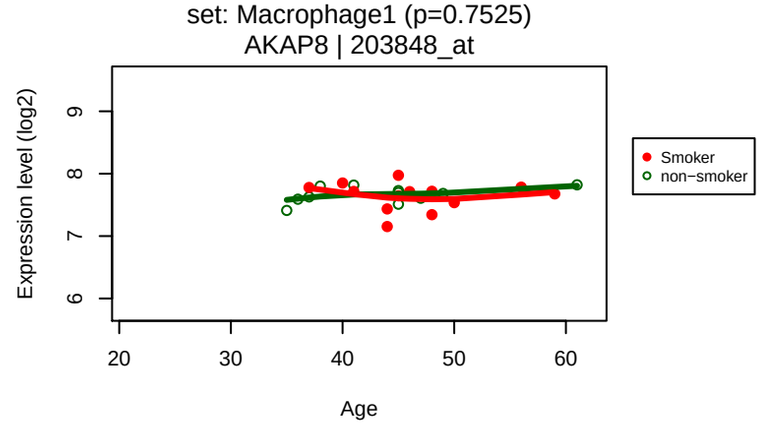

set: SmallAirway1 (p=0.0389)
AKAP8 | 203848_at
Expression level (log2)
Age
Smoker
non−smoker

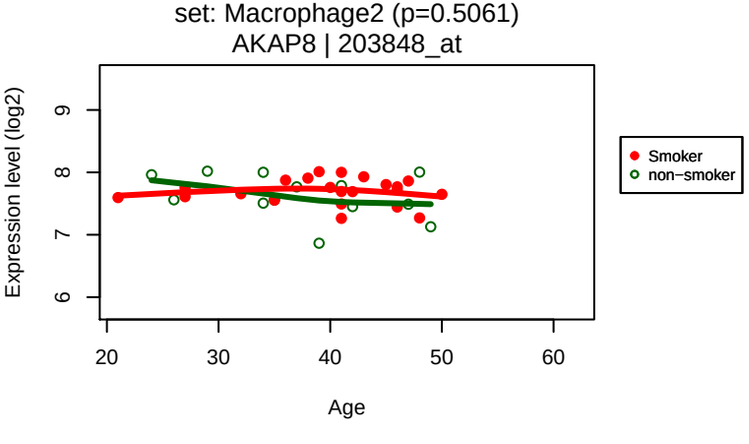

set: SmallAirway2 (p=0.9826)
AKAP8 | 203848_at
Expression level (log2)
Age
Smoker
non−smoker

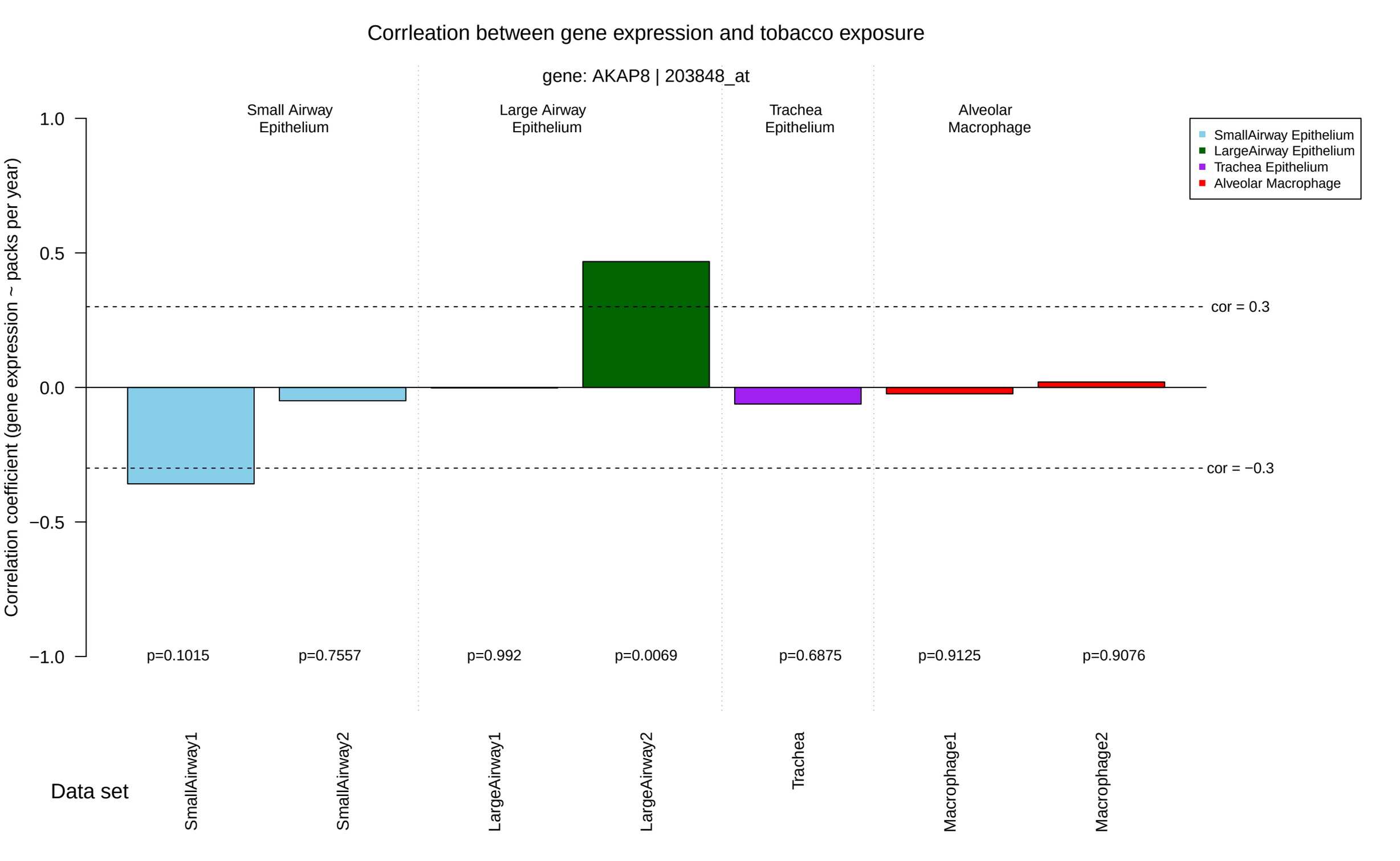

set: LargeAirway1 (p=0.1924)
AKAP8 | 203848_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1848)
AKAP8 | 203848_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5565)
AKAP8 | 203848_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7525)
AKAP8 | 203848_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5061)
AKAP8 | 203848_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP8 | 203848_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1015
p=0.7557
p=0.992
p=0.0069
p=0.6875
p=0.9125
p=0.9076
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2